
From DOWELL SCHLUMBERGER MIDDLE EAST INC, EGYPT
BRANCH SLB Group ("SUBCONTRACTOR")
Branch address at No. 26, Public Free Zone, Ameriya, Egypt

To THE PETROLEUM PROJECTS AND TECHNICAL
CONSULTATIONS CO. ("PETROJET" or "CONTRACTOR")
Office at Plot 21-26, Sector 1, City Center, 5th Settlement, New
Cairo, Cairo, Egypt

SUBCONTRACT No and Date Bridging Agreement No. 194/177-23/AHN/2023
(Including Schedule A and Schedule B)

Letter Date 22/04/2026

Letter Number 000532_EX_313400_LET_SLB_PTJ_1035

Kind Attention Eng. Ehab Ahmed

Subject Notice of Change (VOP Preparation): Concerns over
instructions to deviate from approved Commissioning
Sequence and Plans

Answer Required ☒ YES ☐ NO

Suggested Forward to COMPANY ☒

References 000532_EX_313400_LET_SLB_PTJ_1034 dated 22/04/2026
000532_EX_313400_LET_SLB_PTJ_0971 dated 23/02/2026
000532_EX_313400_LET_SLB_PTJ_0643 dated 12/05/2025

Dear Eng. Ehab Ahmed,

As per Recital E of the Bridging Agreement No. 194/177-23/AHN/2023, DOWELL SCHLUMBERGER MIDDLE EAST INC, EGYPT BRANCH is the sole representative with full power to represent the Subcontractors for all matters, including, but not limited to, issuance of correspondence on behalf of the Subcontractors. Subcontractor/Subcontractors definition is provided in art. 1.1 (b) of the aforesaid Bridging Agreement.

The SUBCONTRACTOR hereby is compelled to record its concerns on the verbal instructions that are being passed on SUBCONTRACTOR's Commissioning team at SITE.

First and foremost, any instruction shall need to be recorded in writing and must be formally received from a CONTRACTOR representative. SUBCONTRACTOR will not take any action until such instructions are formally recorded and discussed.

Secondly, CONTRACTOR GROUP is reminded herewith that SUBCONTRACTOR is operating within the strict framework of the formally approved documents:

1. COMMISSIONING & START-UP SEQUENCE PLAN - document 313400DWPB13239, issued in Final revision on 24/10/2025 following the receipt of Approval by COMPANY on 19/10/2025 with transmittal OUT-3134-A9U-PTJ-00529
2. COMMISSIONING EXECUTION PLAN - document 313400DWPB13235, issued in Final revision on 31/10/2025 following the receipt of Approval by COMPANY on 21/10/2025 with transmittal OUT-3134-A9U-PTJ-00553

However, such received verbal instructions, if executed, would substantially deviate from these approved Plans, causing a material "Scope Creep", which entails the following impacts:

- Unnecessary creation of anticipated Brownfield working conditions in some systems/subsystems, with:
 - o higher safety risks
 - o substantial reorganization of the Isolation and LOTO philosophy, and
 - o extended duration of both Mechanical Completion works (by CONTRACTOR) and Commissioning activities (by SUBCONTRACTOR).
- Inefficient and more costly deployment of Vendors at SITE, with need for extended presence and/or multiple mobilization/demobilization at SUBCONTRACTOR's cost.

For these reasons, SUBCONTRACTOR reserves its rights and puts herewith CONTRACTOR on Notice of Change that, should these deviations from approved Plans be confirmed, SUBCONTRACTOR will:

- Require that its concerns on such deviations are formally recorded in MoM with CONTRACTOR and COMPANY
- Proceed with assessing, compiling, and finalizing all the associated detailed impacts and addressing the same in a VOP according to SCHEDULE A - OFFSHORE SUBCONTRACT SECTION 2. CONDITIONS OF SUBCONTRACT, ARTICLE 14 VARIATION ORDERS and ARTICLE 6 EXTENSION OF TIME".
- Reserve its rights in reference to its exposure to Liquidated Damages as defined in Article 10.2 of Bridging Agreement - 194/177-23/AHN/2023, and waiver thereof as per Article 10(i)-(iii)

Nothing in this letter shall be construed as a waiver or release of rights or otherwise prejudice in any way, any obligations, rights, or remedies which the SUBCONTRACTOR holds under the SUBCONTRACT and/or APPLICABLE LAW, all of which are hereby expressly reserved.

Sincerely yours,

Tamer Mowafy

Tamer Mowafy
Project Manager – SUBCONTRACTOR REPRESENTATIVE

Attachments:

-
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MELEIHA PHASE II – NEW GAS TREATMENT PLANT (EPF)

MELEIHA GAS TREATMENT PLANT (MODULAR EPF)

COMMISSIONING & START-UP SEQUENCE PLAN

EX-DE	05	24/10/2025	Issued For Construction	Peyman S.	Chris P.	Tamer M.		
EX-DE	04	01/10/2025	Re-Issued for Approval	Peyman S.	Chris P.	Tamer M.		
EX-DE	03	04/09/2025	Re-Issued for Approval	Vijay B. / Peyman S.	Chris P.	Tamer M.	-	-
EX-DE	02	31/07/2025	Re-Issued for Approval	Vijay B. / Peyman S.	Chris P.	Tamer M.	-	-
EX-DE	01	02/06/2025	Issued for Approval	Vijay B.	Scott M.	Tamer M.	-	-
EX-DE	00	25/04/2024	Issued for Comments	Vijay B.	Scott M.	Tamer M.	-	-
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 						Contractor Document ID: PF0009-SLB-CO-PLN-0002 Contract N.:		
Petrojet (PTJ) / SLB						Vendor Document ID: NA Purchase Order N.:		
 								
Vendor logo and business name								
NA								
Facility & Sub Facility Description			Project and SoW description			Scale		Sheet of Sheets
3134 & 00 - General			Meleilha Phase II - New Gas Treatment Plant (EPF)			NA		1 / 8
Document Title						Supersedes N.:		
COMMISSIONING & START-UP SEQUENCE PLAN						Superseded by N.:		
						Plant Area NA		Functional Unit 000

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31/07/2025	02	D	NA	Re-Issued for Approval
02/06/2025	01	C	NA	Issued for Approval
25/04/2024	00	B	NA	Issued for Comments
02/04/2024	-	A	NA	Issued for IDC
Date	Client Rev	Contractor Rev	Vendor Rev	Description

HOLD RECORD

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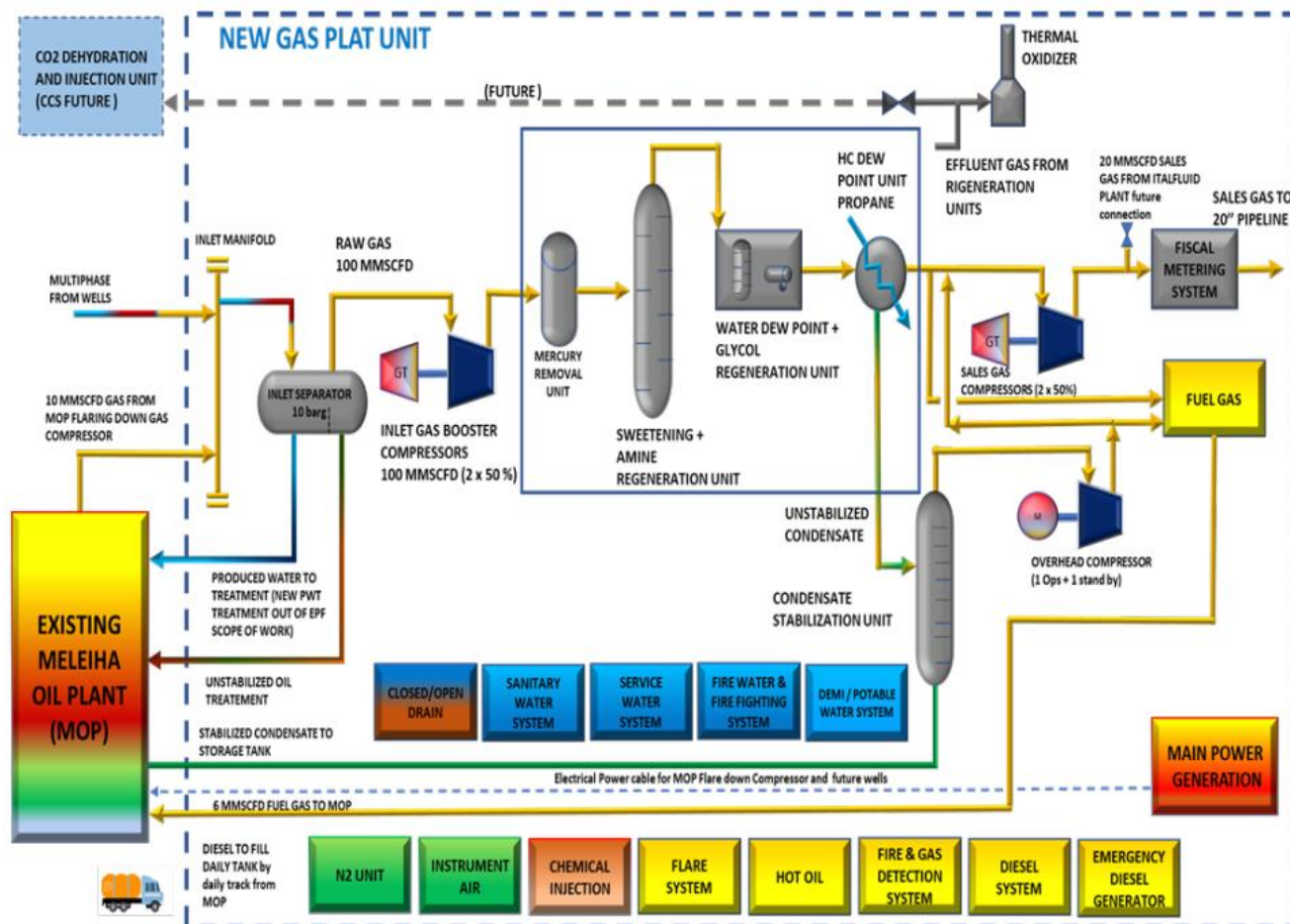
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1.0 PROJECT DESCRIPTION

An overall simplified process scheme is shown in the following block flow diagram.



The new processing facilities receive the collected fields streams from the feed gathering network and treat them in order to obtain the required final sales products. This will require the installation of several treatment units to ensure all the contaminants in the well's fluids are reduced to the targeted level in the facilities products. The facilities will also include all the required utility systems needed to support the process units and achieve the facilities' overall objectives.

The new gas treatment facility shall be able to treat a nameplate capacity for inlet gas of 100 MMSCFD. In addition, the GTP shall be able to receive at inlet a maximum expected HC liquids flowrate of 7000 BOPD as well as a produced water of about 5000 BOPD (to be exported and treated in the existing Meleilha Oil Plant), in order to cover for possible uncertainties on water production (see dedicate section in this document).

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2.0 PURPOSE AND SCOPE

The purpose of this document is to develop a step-by-step commissioning sequence as well as start-up sequence.

This commissioning sequence and logic will be used to develop a system-based commissioning schedule followed by integrating into project Level 3 schedule. Project Mechanical completion schedule then based on this sequence shall be expanded at sub-system level to further develop Level 4 project schedule based on pre-requisites and commissioning/operational needs.

3.0 OBJECTIVE

The objective of this document is to visualize overall sequence and dependencies for commissioning of systems, thereby to help develop logical commissioning schedule.

In addition, this document will support construction and commissioning to easily identify the priorities for construction and mechanical completions. This document also helps Operation teams with regards to visualizing and identifying start-up requirements and readiness, as well as helps to develop detailed start-up sequence.

4.0 DEFINITIONS

Company	AGIBA/ENI
Contractor	Petrojet (PTJ)
Sub-Contractor	SLB
System	A system is a discrete functional section of an installation or facility, either process or utility that performs a major operational function of the plant and can be fully isolated from adjoining systems. The system includes all components/equipment that allows it to operate. The division into systems generally corresponds to the various fluids or services of the installation or facility.
Sub-system	Partial section of a defined System that consists of an independent group of equipment or interconnected items performing a specific function within the pertaining System, which can be isolated and separated from the main associated System to allow for independent Testing and Commissioning.

5.0 ABBREVIATIONS

Term	Definition
FFU	Facility Functional Unit
HVAC	Heating, ventilation, and Air conditioning
ICSS	Integrated control and safety system
MC	Mechanical Completion
P & ID	Process and Instrumentation Diagram
PSUR	Pre-Startup Review
PSUA	Pre-Startup Audit
QC	Quality Control
RFC	Ready for Commissioning
RFH	Ready for Hydrocarbons
RFSU	Ready for Start-Up

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6.0 REFERENCES

Commissioning and start-up sequences were generated in conjunction with below documents as references.

Document No	Client Document No	Document Title
PF0009-SLB-CO-LSL-0001	313400DWEK13240	System List and System/Subsystem Limit Drawings
PF0009-SLB-PR-DBP-0001	313400DPRB12021	Basis of Design - Process
General		All P&IDs including SLB, PRS, and Vendors'.

7.0 COMPLETION SEQUENCE DIAGRAM

Completion sequence diagrams can be referred to as Attachment 1. It comprises of visually representing the systems in order of their commissioning priority. For example, to commission the closed drain system and its respective pumps, power and instrument air systems must be commissioned and available first.

Logic was developed using system/subsystems list and systemization markups, which act as basis to the sequence.

The sequence is divided into several milestones or stages as follows: Temporary power available, ICSS and Instrument Air available, Support utilities available, PSUR Fuel Gas in, RFSU PSUA Hydrocarbon Gas in, and Ready for performance testing / PAC. Further details can be found in Attachment 1.

8.0 START-UP SEQUENCE DIAGRAM

Start-up sequence diagram can be referred to as Attachment 2. The Start-up of the plant will be carried out system wise based on the start-up sequence.

The sequence is divided into several milestones or stages as follows: Pre-requisites / utilities available, Fuel Gas in, Hydrocarbon Gas in, and Process Systems start-up. Further details can be found in Attachment 2.

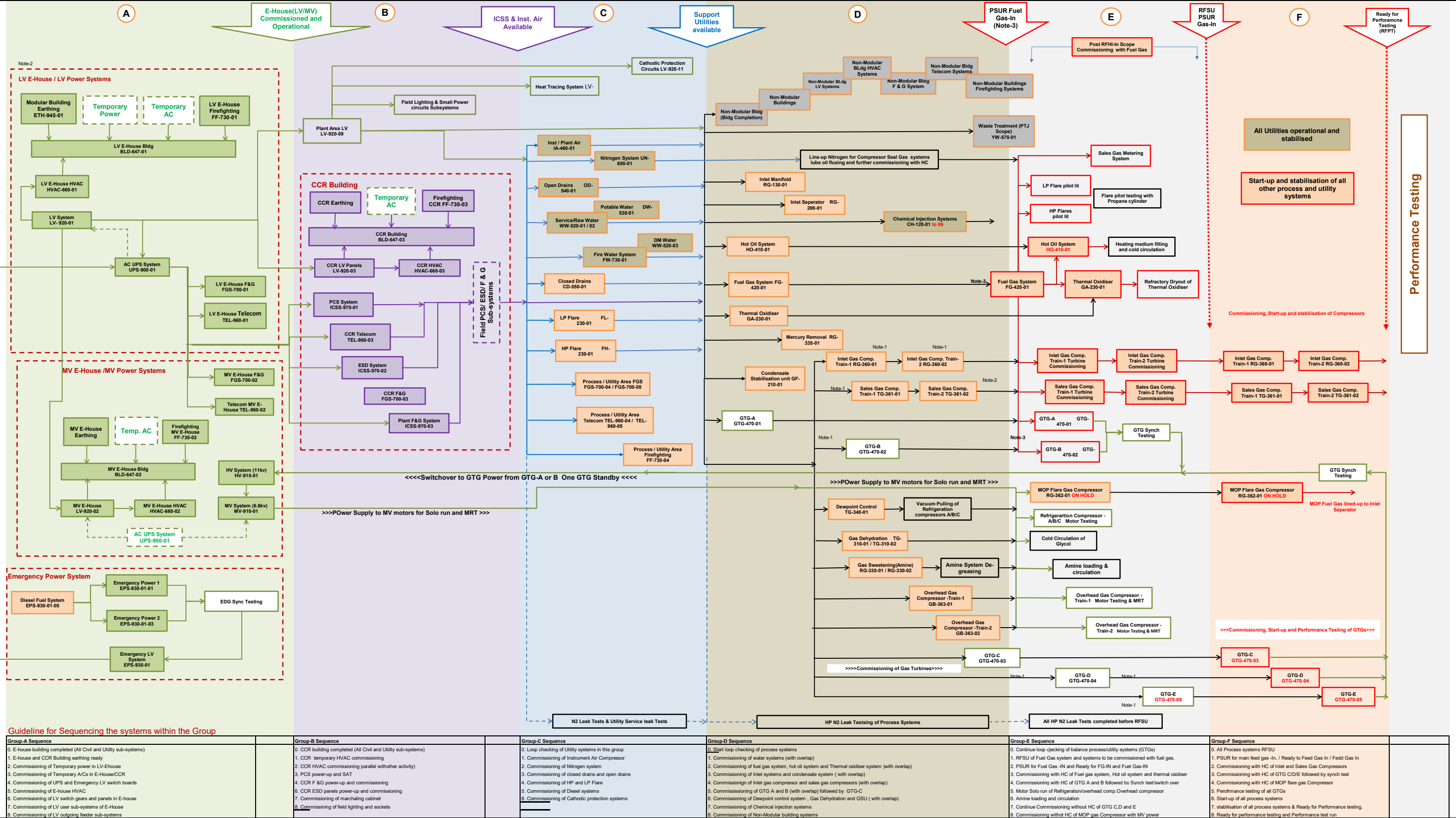
9.0 ATTACHMENTS

9.1 Attachment-1 Completion Sequence Logic

9.2 Attachment-2 Start-up Sequence Logic



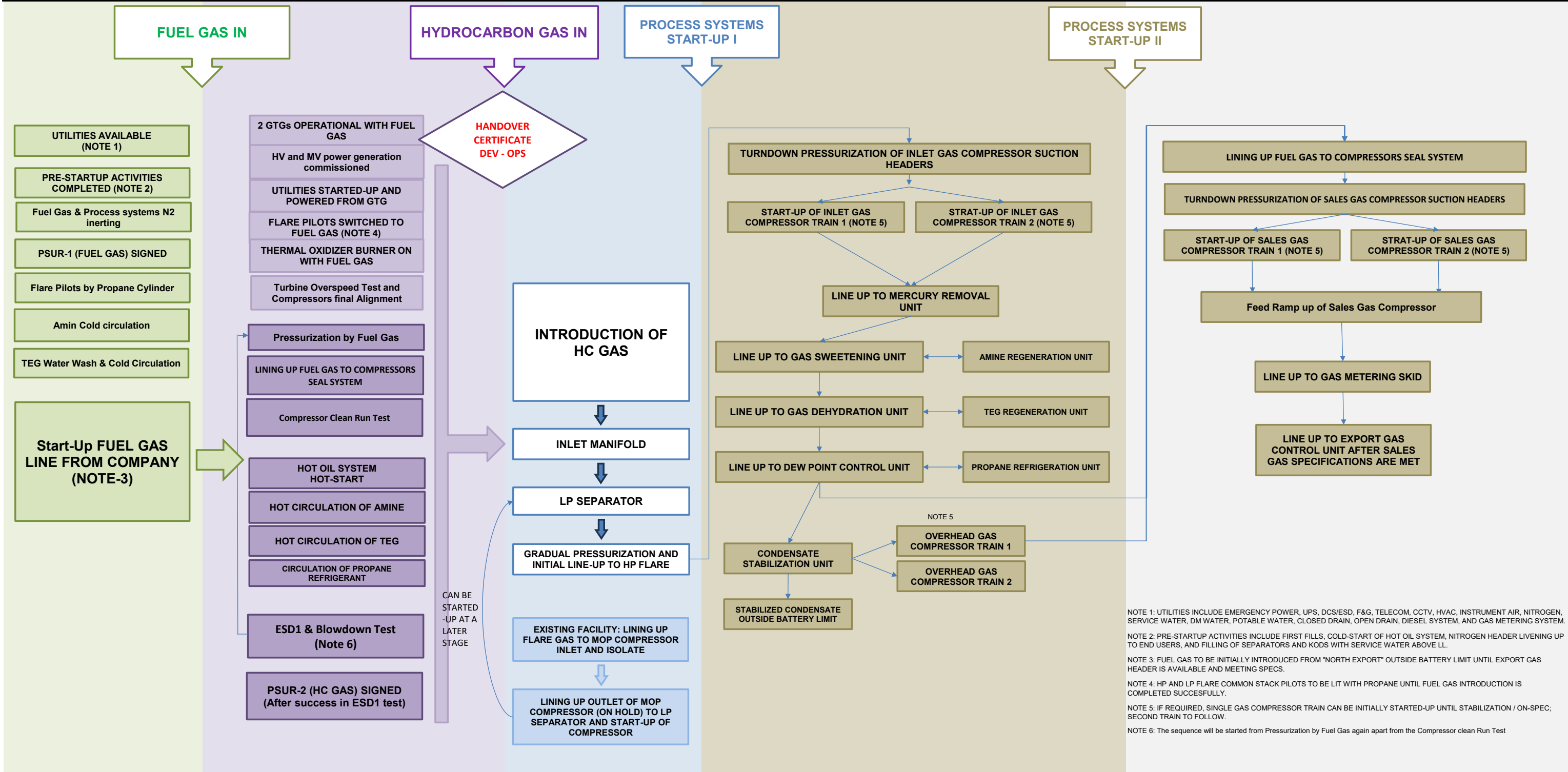
Attachment-1- Completion Logic
COMMISSIONING SEQUENCE LOGIC - Doc.No. PF0009-SLB-CO-PLN-0002
Meleiha Phase II - New Gas Treatment Plant (EPF)



Guideline for Sequencing the systems within the Group

Group-A Sequence	Group-B Sequence	Group-C Sequence	Group-D Sequence	Group-E Sequence	Group-F Sequence
0. E-house building completed (All Civil and Utility sub-systems) 1. E-house and CCR Building earthing ready 2. Commissioning of Temporary power in LV-Ehouse 3. Commissioning of Temporary A/Cs in E-House/CCR 4. Commissioning of UPS and Emergency LV switch boards 5. Commissioning of E-house HVAC 6. Commissioning of LV switch gears and panels in E-house 7. Commissioning of LV user sub-systems of E-House 8. Commissioning of LV outgoing feeder sub-systems	0. CCR building completed (All Civil and Utility sub-systems) 1. CCR temporary HVAC commissioning 2. CCR HVAC commissioning parallel with other activity 3. PCS power-up and SAT 4. CCR F & G power-up and commissioning 5. CCR ESD panels power-up and commissioning 6. CCR ESD panels power-up and commissioning 7. Commissioning of marching cabinet 8. Commissioning of field lighting and sockets	0. Loop checking of Utility systems in this group 1. Commissioning of Instrument Air Compressor 2. Commissioning of Nitrogen system 3. Commissioning of closed drains and open drains 4. Commissioning of HP and LP Flare 5. Commissioning of Diesel systems 6. Commissioning of Cathodic protection systems	0. Start loop checking of process systems 1. Commissioning of water systems (with overlap) 2. Commissioning of fuel gas system, hot oil system and Thermal oxidiser system (with overlap) 3. Commissioning of Inlet systems and condensate system (with overlap) 4. Commissioning of Inlet gas compressor and sales gas compressors (with overlap) 5. Commissioning of GTG A and B (with overlap) followed by GTG-C 6. Commissioning of Dewpoint control system, Gas Dehydration and GSU (with overlap) 7. Commissioning of Chemical injection systems 8. Commissioning of Non-Modular building systems	0. Continue loop checking of balance process/utility systems (GTGs) 1. RFSU of Fuel Gas system and systems to be commissioned with fuel gas. 2. PSUR for Fuel Gas -IN and Ready for FG-IN and Fuel Gas-IN 3. Commissioning with HC of GTG C/D/E followed by synch test 4. Commissioning with HC of GTG A and B followed by Synch test/switch over 5. Motor Solo run of Refrigeration/overhead comp. Overhead compressor 6. Amine loading and circulation 7. Continue Commissioning without HC of GTG C,D and E 8. Commissioning without HC of MOP gas compressor with MV power	0. All Process systems RFSU 1. PSUR for main feed gas -In / Ready to Feed Gas In / Feed Gas In 2. Commissioning with HC of Inlet and Sales Gas Compressors 3. Commissioning with HC of GTG C/D/E followed by synch test 4. Commissioning with HC of MOP flare gas Compressor 5. Performance testing of all GTGs 6. Start-up of all process systems 7. stabilisation of all process systems & Ready for Performance testing. 8. Ready for performance testing and Performance test run

LEGEND	Temporary Supplies	Inst & Control System commissioning	Power Systems commissioning	Utility Systems commissioning without HC	Electrical Field sub-systems	Inst & Control Field sub-systems	Non-Modular buildings (PTJ scope)	Major /Specific Comm. Activity	Process Systems commissioning without HC	Process systems Commissioning with HC	GTG Commissioning with Fuel Gas	Abbreviations	Notes
												PSUR - Pre-start-up review RFSU - Ready for Start-up RFHIN - Ready for Hydrocarbon Introduction	Note-1 Gas Turbine Cranking is part of GTG Commissioning without HC Note-2 Commissioning of Temporary power in LV-Ehouse Note-3 Fuel Gas-In priority is driven by GTG A / B ready for start-up





Project Name

Facility/Subfacility name

Facility/Subfacility code

Country

MELEIHA PHASE II – NEW GTP (EPF)

Melehia Gas Plant

3134

EGYPT

Transmittal out ID

OUT-3134-A9U-PTJ-00529

Date19-10-2025

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Project References:				
From		To		Contract Number
Company	Ahmed Abdelsalam	DC Contractor	Abdelkhalek Yahya	EG20230004
Copy To				

	DOCUMENTATION REFERENCES			REVISION INDEX		TRANSMITTAL IN					
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Related documents

Document	Title
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Raised by: Ahmed Abdelsalam

Project document control

MELEIHA PHASE II – NEW GAS TREATMENT PLANT (EPF)

MELEIHA GAS TREATMENT PLANT (MODULAR EPF)

COMMISSIONING EXECUTION PLAN

EX-DE	04	31/10/2025	Issued for Construction	Vijay B.	Chris P.	Tamer M.		
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 						Contractor Document ID: PF0009-SLB-CO-PLN-0001 Contract N.:		
Petrojet (PTJ) / SLB						Vendor Document ID: NA Purchase Order N.:		
 								
Vendor logo and business name								
NA								
Facility & Sub Facility Description			Project and SoW description			Scale		Sheet of Sheets
3134 & 00 - General			Meleilha Phase II - New Gas Treatment Plant (EPF)			1:1		1/ 35
Document Title						Supersedes N.:		
COMMISSIONING EXECUTION PLAN						Superseded by N.:		
						Plant Area NA		Functional Unit 000

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REVISION LIST

31/10/2025	04	00	NA	Issued for Construction
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28/08/2025	02	D	NA	Re-Issued for Approval
02/06/2025	01	C	NA	Issued for Approval
06/05/2025	00	B	NA	Issued for Comments
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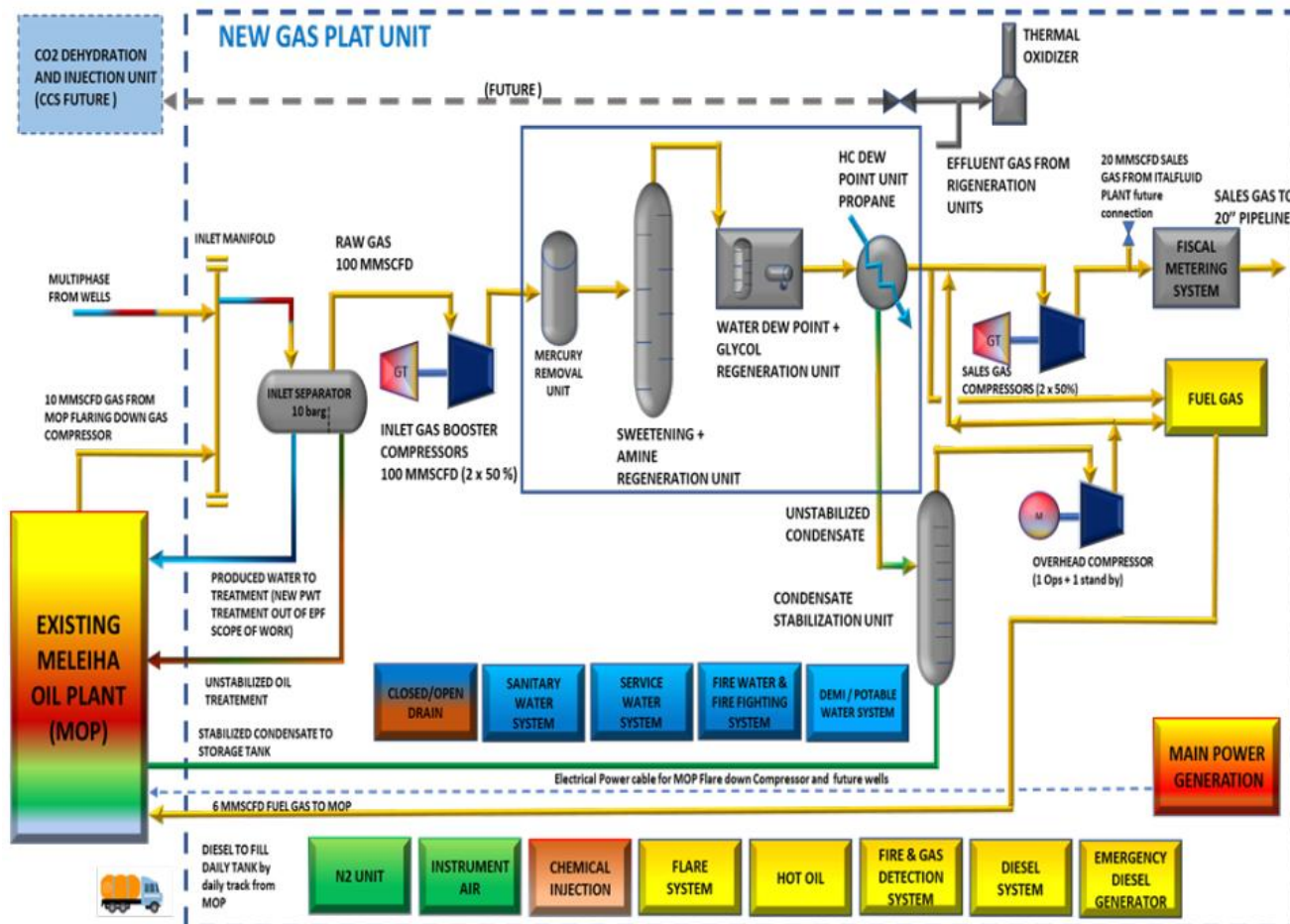
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1.0 PROJECT DESCRIPTION

An overall simplified process scheme is shown in the following block flow diagram.



The new processing facilities receive the collected fields streams from the feed gathering network and treat them in order to obtain the required final sales products. This will require the installation of several treatment units to ensure all the contaminants in the wells fluids are reduced to the targeted level in the facilities products. The facilities will also include all the required utility systems needed to support the process units and achieve the facilities overall objectives.

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2.0 PURPOSE

The purpose of this document is to outline the CONTRACTOR's strategy for executing commissioning of **Maliha Gas Treatment Project (GTP)**, This document shall,

- Provide a roadmap to SUB-CONTRACTOR's key completions management controls and procedures required to meet the project specifications and thereby achieving smooth commissioning and start-up of the facility.
- To ensure that this commissioning execution plan is clearly understood and shared from early stage such that the preparation / execution work is conducted for Safe delivery of the physical assets to the Company.
- To ensure clarity and understanding as to the demarcation of COMPANY, CONTRACTOR's, and SUB-CONTRACTOR's scope of responsibility pertaining to commissioning and start-up.
- Develop and execute "static and dynamic Commissioning activities to prove the functionality of the systems and associated equipment with or without the introduction of hydrocarbons, partial and/or temporary energizing and pressurizing of a system.
- Develop and execute the commissioning schedule/plan to achieve a successful commissioning, start-up, initial operation, and performance testing of the systems.
- To ensure that the Contractual expectations are met via procedures, commissioning and completions management systems, system handover dossiers and performance tests logs.
- Outline the SUB-CONTRACTOR's QC and HSE principles associated with mechanical completion and commissioning.
- Outline the SUB-CONTRACTOR's resources management plan, including vendor representative, third parties and temporary equipment.

3.0 SCOPE

This plan will describe the sub-contractor's scope associated with commissioning during the following phases of the project:

- Engineering/Procurement Phase (Home Office Preparation and design review Phase)
- Mechanical Completion (including Pre-Commissioning) (Outlining the Contractor's scope of Mechanical Completion)
- Commissioning, Start-up, Stable Operations and Performance Testing phase, as per CONTRACT and project specifications.

This commissioning execution plan shall further align with CONTRACT between contractor and sub-contractor, the company specifications and the project specific procedures as referred in Section-6 References.

4.0 DEFINITIONS AND ABBRIVIATIONS

Company	AGIBA/ENI
Contractor	Petrojet (PTJ)
Sub-Contractor	SLB
CCMS	(Completion & Commissioning Management System) An automated, computer aided Database used for Management, data storage, control and reporting of Construction, Mechanical Completion and Commissioning processes.

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Mechanical Completions (MC)	The progress status achieved when fabrication, erection, installation, integrity testing, cleaning, re-instatement and preservation of all single devices, assemblies and equipment included in the Sub-System have been completed and documented in accordance with specified design requirements.
Commissioning	Means the systematic process aimed to orderly execute and document, on a System and Sub-System base, agreed functional, service and OPERATIONAL TESTs necessary to bring the EPF from the construction stage up to START UP and stable operation, in compliance with approved design requirements and applicable rules, regulation, codes and standards
Pre-start-up Review (PSUR)	Means the auditing session, jointly conducted by COMPANY and CONTRACTOR aimed to ascertain the EPF Facility's readiness for the safe introduction of well's Hydrocarbons or First Charge.
Handover	Transfer of Facility's portions based on a Sub- System/Systems determination between functional groups within Contactor's or Sub-contractor's Organization (i.e.: Construction, Commissioning, Operations & Maintenance Organizations, etc.).
System	A system is a discrete functional section of an installation or facility, either process or utility that performs a major operational function of the plant and can be fully isolated from adjoining systems. The system includes all components/equipment that allows it to operate. The division into systems generally corresponds to the various fluids or services of the installation or facility.
Sub-system	Partial section of a defined System that consists of an independent group of equipment or interconnected items performing a specific function within the pertaining System, which can be isolated and separated from the main associated System to allow for independent Testing and Commissioning.
Punch Item	An incomplete, defective, damaged, poorly executed, or poorly preserved part of work or an equipment malfunction, associated with a component or assembly forming part of a Sub- System or System, requiring additional work or re-work to be resolved., with mutually agreed exceptions to carry over to the next phase.
Ready for Commissioning RFC)	The completion of construction and Mechanical Completion by way of verification of A ITR's for the relevant sub-systems and acceptance by Company with category B Punch List with exceptions to carry over to commissioning phase.
Ready for Hydrocarbon-In (RFHC-IN)	Means the completion status of the EPF achieved when the EPF is ready for the introduction of well hydrocarbons, the Ready For Start-Up Certificate have been Approved and issued by COMPANY for all Systems and the PSUR have been successfully completed.
Ready for Start-up (RFSU)	Means the completion status achieved by a System when all related Commissioning activities, to be performed before the introduction of well hydrocarbon (i.e.: Functional Tests, Operational tests, Preparatory activities for the Start-Up, etc.), have been completed and documented with exceptions to carry over and complete remaining ITR/Punch list during operations phase.
Provisional Acceptance Certificate (PAC)	PAC is issued on successful completion of performance test run of the facility as per performance guarantees specified in the project Performance testing procedure
Final Acceptance certificate (FAC)	FAC is issued on closure of all warranty and defect issues at the end of contractual warranty period

Term	Definition
RFSU	Ready for Start-up
RFC	Ready for Commissioning

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MC	Mechanical Completion
QC	Quality Control
FAT	Factory Acceptance Test
HSE	Health Safety & Environment
PSUR	Pre-start-up Review
ITR	Inspection Test Record
PTW	Permit to work
CSU	Commissioning and start-up
PSUA	Pre-start-up Audit
RFI	Request for Inspection
HVAC	Heating Ventilating and Air Conditioning
ESD	Emergency Shut Down
ICSS	Instrumentation & Controls Safeguarding System
UPS	Uninterrupted Power Supply
DCS	Distributed Control System
F&G	Fire & Gas
SAT	Site Acceptance Test
CCMS	Completion and Commissioning Management System
CMMS	Computerized Maintenance Management system
LOTO	Lockout Tagout
HAZOP	Hazardous Operations
SIMOPS	Simultaneous Operations
MSDS	Material Safety Data sheet
KPI	Key Performance Indicators
CTP/OTP	Commissioning Test Procedure / Operational Test Procedure
IFC	Issued for Construction
MDL	Master Document List
P&ID	Piping & Instrumentation Diagrams
SLD	Single Line Diagram

5.0 REFERENCES

This document is developed in-line with the scope of work associated with construction, mechanical completions and commissioning requirements described in following documents from ITT.

Document No	Document Title
313400FGSC0103	SCOPE OF WORK FOR GAS-TREATMENT PLANT (EPF) EPCC
Attachment-2 of above SOW	Completion and Commissioning Strategy REV.00
27952.COS.AVV.STD	Facilities Commissioning Specification
27950.COS.QUA.STD	Facility Completion
27951.COS.QUA.STD	Facility Construction, QC, and Mechanical Completion Specification
27970.COS.AVV.STD	Commissioning Inspection Test Plan & Inspection Test Reports
PF0009-SLB-CO-PRO-0001	Completions, Acceptance and Handover Procedure
PF0009-SLB-CO-PRO-0002	CCMS implementation procedure
PF0009-SLB-CO-PLN-0002	Commissioning and Start-up Sequence plan

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6.0 ROLES AND RESPONSIBILITIES

6.1 COMPANY(AGIBA/ENI) AND CONTRACTOR (PETROJET)

- Approval of key commissioning deliverables as described in this document, as may be applicable by CONTRACT and approved MDL.
- Review and approve system/sub-system list
- Review and approve commissioning deliverables as per the category assigned in approved MDL
- Review and approve project commissioning schedule.
- Review sub-systems Mechanical Completions submitted by contractor.
- Approve the punch list proposed by CONTRACTOR for closeout.
- Attend punch list consolidation meeting after MC walkdowns.
- Approve system completion dossier with outstanding Category-C punch lists.
- Review and approve system commissioning dossiers submitted by sub-contractor.
- Review and approve plant performance test procedure.

6.2 SUB-CONTRACTOR (SLB)

- Population of the Technical Database into Completion commissioning management system (CCMS)
- Approval of third-party vendor's method statements for specialist commissioning activities like preservation, leak testing, lube oil system flushing, rotating equipment alignment, chemical cleaning etc.
- Develop CSU deliverables as listed in this document or as per CONTRACT requirements.
- To identify requirements for temporary utilities and equipment and manage the mobilization and operation at site.
- Review Mechanical Completion (MC) dossiers prior to acceptance for onwards commissioning.
- Prepare and submit system completion dossiers for company/contractor review and approval.
- Manage the Vendor assistance during the Construction, Commissioning, start-up and performance testing phase of the project.
- Review vendor's preservation specifications, commissioning procedures and IOMs.
- Oversee Preservation activities executed by contractor preservation team.
- Identify First Fills and prepare first fills (as per CONTRACT) scheduled with necessary details like type, quantity, MSDS and filling schedule. Perform the first fills as per the mechanical completion schedule.
- The sub-Contractor (SLB), along with the Company, reviews all Punch Item categories (especially "B" and "C" items) before being handed over the sub-Systems

6.3 SUB-CONTRACTOR'S COMMISSIONING MANAGER

Sub-CONTRACTOR's Commissioning Manager will be responsible for,

- Completion monitoring/prioritization and Commissioning preparation work
- Manage commissioning interfaces with internal stakeholders, client, and other SUB-CONTRACTORS.
- Manage Commissioning planning and scheduling including integration with construction team for prioritizing construction activities in line with the commissioning priorities
- Oversee Health, Safety & Environment implementation during commissioning, start-up, and initial operation phase. Define PTW authorities during the Commissioning phase, enforcing adherence to

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PTW system, to control the SIMOPs between Construction, Commissioning and Operations/Maintenance. Coordinating the transition from the Construction PTW system, prior to energization and/or hydrocarbon introduction.

- Planning and managing mobilization of commissioning resources, third parties and vendor specialists
- Organize Training of company personnel as per project scope of work
- Hold regular review meetings with the client on issues such as HSE conformance and commissioning progress as well as meetings with the vendors to resolve any equipment related problems.
- Review reports generated by Commissioning Engineers
- Provide support and technical guidance to the commissioning team, as and when required
- Lead PSUR activities and monitor action closeouts.

6.4 SUB-CONTRACTOR'S COMMISSIONING COORDINATOR

The sub-contractor's Commissioning coordinator shall be responsible for following.

- Reports to the Commissioning Manager
- Acts as a Commissioning focal point during the internal or external handover processes.
- Inform Commissioning Manager of status, activities, and other relevant Pre-commissioning and commissioning activities.
- Supports the development of system definitions, pre-commissioning/Commissioning procedures, and schedules that are aligned with Client's and Project requirements.
- Assists in the development of the mobilization plans and organization charts.
- Attends construction and project meetings that are pertinent to his area of responsibility.
- Prepares the weekly progress report and submits it to the Commissioning Manager
- conducts daily meetings with the Commissioning Team.
- Assigns commissioning tasks, objectives, and targets to the Commissioning Team. Liaise with COMPANY Commissioning and Operation Teams, VENDOR representatives and project engineering team as required.
- Liaise with third party services vendors (e.g. Leak testing, Lube oil flushing, Rental equipment vendors) to ensure timely mobilization and oversee their activities.

6.5 SUB-CONTRACTOR'S COMMISSIONING DISCIPLINE LEADS

- They are discipline (e.g., Electrical, Instruments, Mechanical, etc) experts within the commissioning team. They will be Responsible for supervising the Commissioning Discipline engineers and supervisors during commissioning phase. They shall ensure that the relevant plant commissioning procedure(s) and VENDOR procedure(s) are followed during commissioning.
- Ensure commissioning test records (B-ITRs) are completed per system or sub-system and approved ITRs are fed to CCMS engineer to update the progress into CCMS.
- All commissioning activities within their respective discipline, which require Vendor participation, will be under the control and direction of the (Discipline) commissioning lead.
- To liaise with CONTRACTOR for their scope of Pre-commissioning and assistance require during commissioning.
- Consolidate punch lists items with its category (A or B) in consultation with company authorities, commissioning team and submit the final list to CCMS engineer.
- Expediting the liquidation and closing of punch list items under responsibility Contractor.

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- Issue the material request and follow-up the procurement progress for items requires closing the punch list.
- Responsible for signing off and collecting the copy of the vendor's Field commissioning report and verify that vendor's work has been completed. He also ensures that vendor reports are suitably archived, and copies are filed in respective dossiers.
- Responsible for validating and signing off the Vendors' time sheet, and control system Vendor's costs in line with the budget.

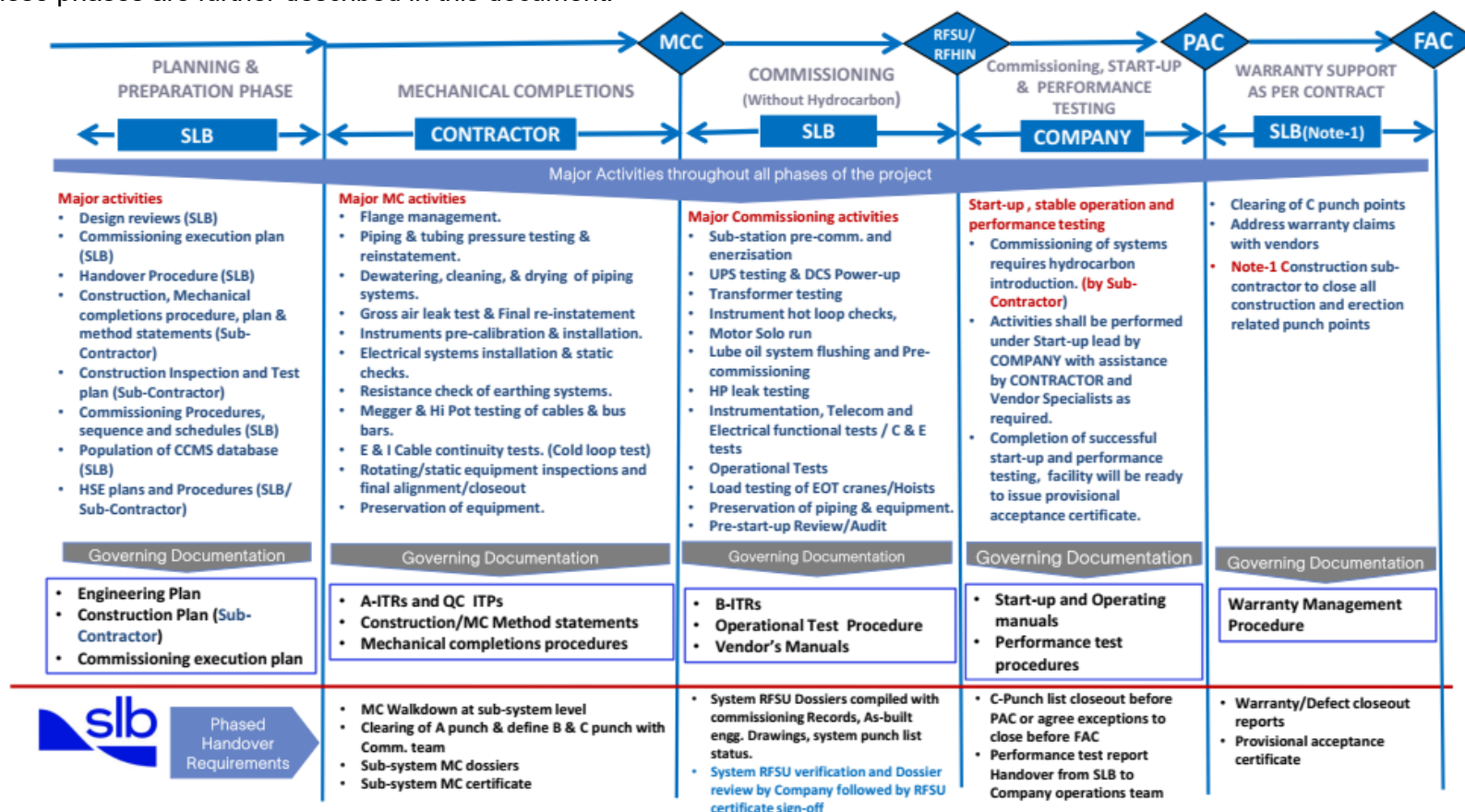
6.6 SUB-CONTRACTOR'S COMPLETIONS ENGINEER (CCMS ENGINEER)

- Map/configure systems and sub-systems into CCMS.
- Coordinates with commissioning team and collects all required data for CCMS.
- Import/populate tag data to CCMS application and confirm all construction and Pre-commissioning check sheets are assigned correctly.
- Update the approved check sheets to CCMS.
- Generate sub-system MC certificate on completion of all A and B ITRs within a sub-system.
- Generate system completions certificates on completion of all commissioning activities at system level and leak test pack level.
- Scan and save all commissioning check-sheets under respective systems and sub-system.
- Update the Punch lists as directed by the Punch list coordinator/discipline commissioning lead.
- Keep record of Field Engineering queries and MOC associated with commissioning.
- Prepare MC dossier and system completion dossier as per requirement described in this document.
- Prepare weekly progress report as per commissioning priority; also Prepare monthly progress report for the construction and commissioning status.

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7.0 COMPLETIONS AND COMMISSIONING SCOPE SPLITS V/S PROJECT PHASES

Following chart gives overview of major activities, scope split, documentation and governance involved from planning and preparation phase through MC, Commissioning, start-up, performance testing and warranty phase of the AGIBA project. Detailed activities through all these phases are further described in this document.



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8.0 COMMISSIONING PLANNING AND PREPARATION

Sub-Contractor's Commissioning Manager or head office commissioning specialist will be involved early during the Engineering phase of the project to provide input to Engineering to address commissioning requirements into the design, to review vendor's CSU procedures including FAT Procedures and may attend FATs as necessary.

As part of project completion and commissioning planning, following key activities shall be carried out by commissioning manager and commissioning specialist from HQ operations support team,

8.1 ENGINEERING AND PROCUREMENT INPUT AND DESIGN REVIEWS.

Commissioning Manager and engineers shall involve in the following major activities during Engineering and procurement phase of the project including early planning and preparation of commissioning deliverables.

- Review and comment on engineering drawings, e.g., P&ID's, PFD's, SLDs, Instrument & Control drawings, ICSS Architecture etc., and participate in HAZOP, SIL Reviews.
- Participate in 3D model reviews (30%,60%& 90% reviews), to input the requirement for commissioning & operational inputs.
- Define Systems / Sub-Systems lists and boundaries (e.g., P&ID/SLD/Layouts markups)
- System / sub-system activities list; identifying the requirement for commissioning activities.
- Review FAT, SAT procedure and schedule.
- To set-up and populate Commissioning and Completions Management system (CCMS) based on the system / sub-system list and tag data.
- Support Project control: Prepare the activity list, tasks & sequence of pre-commissioning and commissioning, and provide support to Project control and planning team to develop integrated project schedule.
- Development, review and approval of commissioning, Start-Up, and testing procedures.
- To review Mechanical completions and pre-commissioning procedures and method statements developed by Contractor.
- Define scope of the work for specialized commissioning services such as special cleaning and leak testing / activities, identify required special tools & temporary equipment's.
- sub-contracting of specialized third-party commissioning services as applicable to the project.
- Coordinate Procurement of required tools, test equipment and consumables for commissioning.
- Develop and implement the preservation plan, and set-up preservation matrix.

8.2 Commissioning and Completions Documentation

Following documents, plans, procedures, registers, and check lists are envisaged for smooth and systematic execution and completion of this project.

8.2.1 Commissioning and Completions Management plans and procedures (Deliverables)

The following documents are key to a smooth and systematic execution and management of commissioning and completion of this project.

1. Commissioning execution plan (This document)

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2. Completion, Acceptance and Handover Procedure. (this includes Punch Management)
3. CCMS Implementation Procedure
4. Commissioning Organization chart and Commissioning Manpower mobilization plan
5. Vendor mobilization plan
6. Commissioning Sequence (Input to Project L3 / L4 schedule and to develop MC skyline)
7. System list and Sub-Systems mark-up drawings
8. A and B ITRs catalogues and Tag type allocation matrix

8.2.2 Commissioning HSE plans and procedures

The following procedures are key to safe execution commissioning of this project.

1. PTW Procedure (Review/Input by commissioning team)
2. Energization Procedure (Including LUN)
3. Lock out Tagout Procedure
4. Spade and Blind control Procedure
5. Control of Temporary links & Software Inhibits / Overrides Procedure.

8.2.3 Mechanical Completion (Including Pre-Comm) Procedures (Contractor Scope)

MC and Pre-commissioning procedures shall be developed by contractor (Petrojet) with technical assistance by sub-contractor. These procedures typically include:

1. MC and Pre-Commissioning Procedure- Mechanical and Piping
2. MC and Pre-Commissioning Procedure – Electrical, Instrumentation and Telecom

8.2.4 Pre-Commissioning and Commissioning General procedures

1. Loop Testing Procedure
2. ICSS, ESD/F & G Block, Force & Override procedure
3. Leak Testing and inerting Procedure
4. Chemical cleaning Procedure (or any special cleaning as per vendor recommendation)
5. Lube oil system flushing procedure and mark-up drawings.

8.2.5 Pre-commissioning and Commissioning Registers

The following registers shall be maintained through the commissioning phase and shall be handed over to company along with the system dossiers.

1. First Fills Register
2. Temporary links & Software Inhibits / Overrides register.
3. Spade and Blind Register or Process Isolation register
4. Special equipment Register (Supplied by equipment vendors)
5. Lock open / Lock closed (LO/LC) valve Register.
6. PSV Testing Record (PSV Register)

8.2.6 Commissioning and operational testing Procedures

Commissioning and start-up shall be executed using step-by-step procedures as listed below. In case of a small system or small skid package, commissioning ITR-B shall be updated in-line with vendor's commissioning procedure and industries common practice.

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For commissioning complex systems involving multiple checks and tests, system wise Operation test procedures (OTP) shall be developed. OTPs identified for this project based on the system list, are as below:

1. Utility Commissioning Procedures (OTP)
A service leak test of the utility systems will be included in the relevant utility system's OTP.
2. Process commissioning procedures (OTP)
This Includes all process system OTPs for non-hydrocarbon commissioning scope and hydrocarbon commissioning scope. OTPs for the Systems which require commissioning with hydrocarbon shall be developed separately, or alternatively an Appendix for HC commissioning shall be added to the main OTP for such systems.

Refer Annexure-1 for Pre-Commissioning, Commissioning, start-up and performance testing deliverables and Annexure-2 for List of Operational Test procedures (OTPs)

3. Performance Test Procedures
This procedure shall be developed to ensure that process guarantee requirements are properly covered. Project Contractual performance guarantee is the key basis of this procedure.

8.2.7 Start-up, shut down and operations documents

The following documents shall be developed in-line with the start-up sequence, operating philosophy, vendor IOMs and key engineering documents such as Control Narratives and Cause and Effects matrix.

1. Start-up and shutdown procedure
2. Plant Operating procedure Manuals (as per project scope)

9.0 COMPLETIONS AND COMMISSIONING MANAGEMENT SYSTEM(CCMS)

The completion activities of a **Maliha Gas Treatment Project (GTP) Project** consists of a series of quality checks on the build of the plant which continues through mechanical completion and commissioning phase. Sub-Contractor will utilize their Completions and Commissioning Management System (CCMS), which is built under Maximo platform. Data within Maximo CCMS can be also used as CMMS (Computerized Maintenance Management System) for planning and tracking maintenance activities post start-up.

MAXIMO CCMS will provide the following features from construction through the commissioning phase of the project:

- Mapping of system and sub-system hierarchy
- Tag data population and assigning system and sub-system to all the tags.
- Construction A-ITR and Commissioning B-ITR allocation as per ITR allocation matrix.
- Generate the ITRs per sub-systems.
- Scanning and uploading the completed ITRs into the CCMS system and thereby monitoring the sub-system and system completion progress.
- Punch list assignment against system and sub-system
- Punch list monitoring and reporting

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- By linking the CCMS with Power BI, facilitate issuing status reports of individual system and subsystems allowing the construction and commissioning to drive the construction and commissioning priorities.
- Certificates are generated based on status of system and sub-system completion as below,
 - Mechanical completion Certificate (MCC)
 - Ready for Startup Certificate (RFSU)
 - Ready for Hydrocarbon-In certificate (RFHIC)

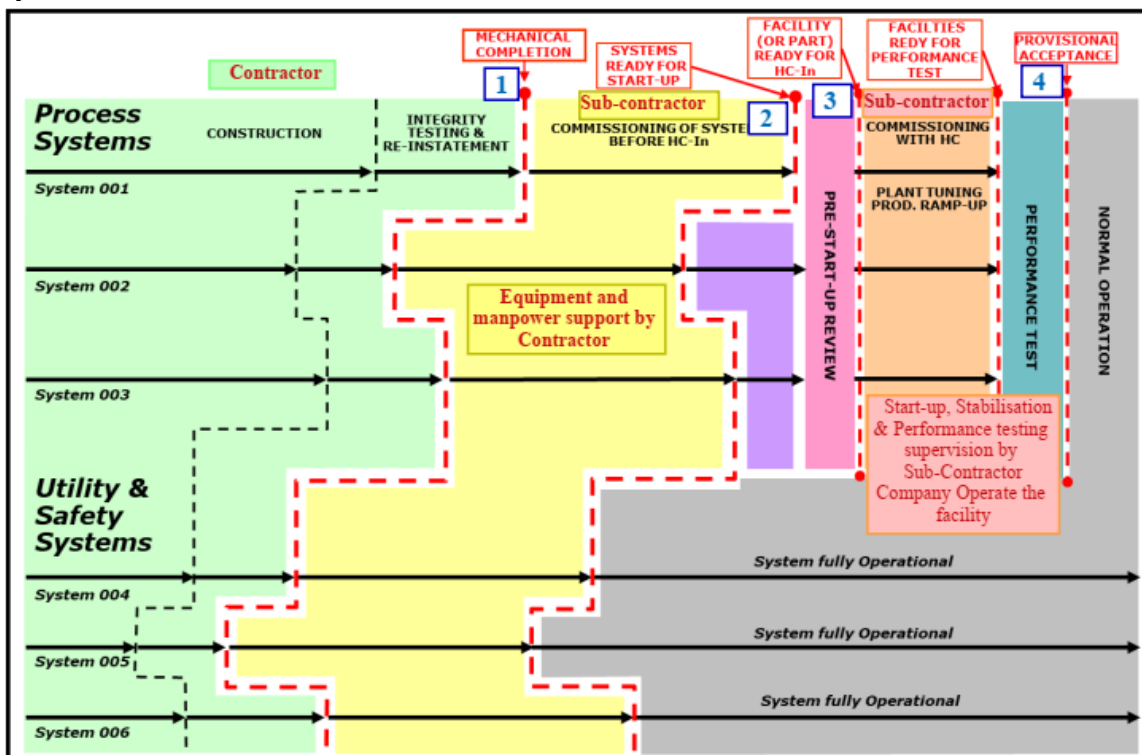
Project CCMS Implementation is further detailed in Document: **PF0009-SLB-CO-PRO-0002 CCMS implementation procedure**

10.0 PROJECT COMPLETIONS AND HANDOVER STRATEGY

Major milestones for completion and handover of **Meleiha Gas Treatment project** in line with project specification 27951.COS.QUA.STD Facilities Construction Quality Control and Mechanical Completion Specification are as below.

1. Mechanical completions phase
2. Commissioning Phase
3. Facilities start-up and Performance Testing Phase
4. Provisional Acceptance and Operation Phase

Following chart from the project specification 27951.COS.QUA.STD shall be followed for completions and handover of the facility between contractor and sub-contractor as well as between contractor and company.



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Completions, Acceptance and handover process is further described in *Completions, Acceptance and Handover Procedure (PF0009-SLB-CO-PRO-0001)*. This document covers following details,

- Scope split between contractor and sub-contractor
- Key milestone completion requirements (e.g. MC, Commissioning and RFSU)
- Completion walkdowns and Punchlist Management
- Completions dossiers review and approval process (MC and system completion dossiers)
- All the templates for MC and System dossier Index
- All templates for completion certificates as below,
 - ✓ MC Certificate
 - ✓ RFSU Certificate
 - ✓ RFHCIn Certificate
 - ✓ RFPT Certificate
 - ✓ Provisional Acceptance and Final Acceptance certificates

11.0 COMMISSIONING EXECUTION STRATEGY

This section of commissioning execution plan describes scope of commissioning systems, commissioning QC and HSE principles, commissioning sequencing and planning, commissioning/start-up/performance testing methodology, scope of commissioning before and after introduction of hydrocarbon, start-up and performance testing.

Sub-contractor considers commissioning of a system in below hierarchy and priority

1. Installation checks through A-ITR will be supplemented with the vendor's SAT procedure checklists for installation checks, wherever vendor has provided SAT procedure.
2. Commissioning checks through B-ITRs will be supplemented with the vendor's SAT procedure checks for functional tests and checks wherever vendor has provided SAT procedure
3. Vendor's SAT procedure will also be used to ensure that necessary functional tests are covered in OTP, thereby vendor's SAT checklists can be endorsed by refereeing relevant OTPs.

11.1 COMMISSIONING SYSTEMS/UNITS

Commissioning and completions systemization and commissioning sequence will be developed based on following main systems/units of **Meleiha Gas Treatment project** broadly divided to have overview of commissioning & completion scope. These systems/units are further divided for step-by-step completion and commissioning of the Meleiha GTP facility, into a detailed systems and sub-systems list.

A. Hydrocarbon Process Systems	B. Process Support systems
1. Inlet manifold 2. Inlet separator 3. Inlet gas compression- Train 1 4. Inlet gas compression - Train-2 5. Mercury removal unit (MRU) 6. Gas sweetening unit (Amine regeneration) 7. Gas sweetening unit (Amine closed drain) 8. Gas dehydration unit (TEG)	18. Corrosion Inhibitor System 19. Demulsifier Injection System 20. Antifoam Injection System 21. Scale Inhibitor System 22. MEG Injection System 23. Sodium Hypochlorite System 24. Hot oil system 25. HP flare system

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9. Gas dehydration unit (Glycol closed drain) 10. Dew point control unit 11. Overhead gas compression train 1 12. Overhead gas compression train 2 13. Sales gas compression train 1 14. Sales gas compression train 2 15. MOP flare gas compression 16. Sales gas metering & export 17. Condensate Stabilization System	26. LP flare system 27. Thermal oxidizer system 28. Fuel gas system
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C. Utility Systems	D. Electrical, Instrumentation, Telecom & HVAC
29. Raw water system 30. Service water system 31. Demineralized water system 32. Potable water system 33. Closed drain 34. Open drain 35. Diesel fuel system 36. Compressed air systems (Inst. Air) 37. Nitrogen system 38. Waste treatment and disposal (Petrojet) 39. Fire water systems	40. HV Power Systems (11KV) 41. MV Power Systems (6.6kv) 42. LV Power Systems 43. UPS Systems 44. Emergency Power Systems 45. Earthing systems 46. Power Generation Systems 47. Control Systems (PCS/ESD) 48. Fire and Gas systems 49. Telecom Systems 50. Building HVAC systems

Refer to the *Systems and sub-systems mark-up and system list* (PF0009-SLB-CO-LSL-0001) for detailed overview of the systems and sub-systems applicable for this project.

11.2 TRANSITION FROM MECHANICAL COMPLETION TO COMMISSIONING

11.2.1 Commissioning Pre-requisites

Commissioning Manager and commissioning leads to ensure that following requirements are met prior to energizing/pressurizing and commence commissioning of the systems

- a. Module/skid fabrication, equipment installation, integration and FAT at vendor's shop are completed and following documentations are available,
 - ✓ Factory Acceptance tests reports
 - ✓ Outstanding punch lists report with all category-A punch items are closed
- b. Site Mechanical completion of systems and sub-systems
 - ✓ All applicable ITR-A are filled
 - ✓ Contractor (Petrojet) has completed all site installation, hook-up and pressure testing as applicable.
 - ✓ Commissioning team has attended critical system's walkdowns and reviewed MC dossiers to ensure completeness
 - ✓ Sub-systems Mechanical completion achieved MC certificate issued

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11.2.2 Mechanical Completion scope (Petrojet)

Mechanical Completion is a final phase of construction activities to verify the completeness of the plant that each installed component conforms to or is fabricated, installed and checked in accordance with the project specifications and procedures

For a system, sub-system or equipment, Completion of following activities are minimum for achieving Mechanical completion with associated ITR-A are signed at project construction site.

- Static equipment/packages: Internal cleaning, inspection and filled with catalyst (if any).
- Rotating equipment: All installation work is completed, and final alignment signed.
- EIT: All terminations and construction checks/continuity checks are completed.
- Piping: Post-hydro test de-watering, drying and interting are completed and witnessed/signed by commissioning.

11.2.3 Mechanical Completion plan and procedures

Contractor (Petrojet) shall develop *Mechanical completion plan and associated procedures* for site construction and mechanical completions scope and shall be reviewed by sub-contractor QC and Commissioning team as applicable. The contractor shall consider following project specifications and project specific procedures while developing Mechanical Completion plan and procedures.

- ✓ 27950.COS.QUA.STD Facility Completion
- ✓ 27951.COS.QUA.STD Facility Const, QC, and Mechanical Completion Specification
- ✓ 27970.COS.AVV.STD Commissioning Inspection Test Plan & Inspection Test Reports
- ✓ PF0009-SLB-CO-PRO-0001 Completions, Acceptance and Handover Procedure

11.2.4 Mechanical Completion & Pre-Commissioning QA/QC Principles

Contractor's Quality Assurance and Quality control team, during Mechanical Completions and Pre-commissioning process shall be responsible for,

- Development of ITPs (Inspection and Test plans) for construction and mechanical completions using inspection and test records (ITRs-A) for verifying the erection and installation activities
- Development of ITR-A matrix for various inspection activities during construction phase
- Quality control procedures for various Mechanical completion activities as below
 - ✓ Hydrotest
 - ✓ Flushing, cleaning and drying.
 - ✓ E & I installation
 - ✓ ATEX/Ex Inspection plan

11.2.5 Punchlist Management during construction to commissioning transition

Prior to the transfer of responsibilities for a System/Sub-system from MC to Commissioning, a tri-party walkdown shall be done to verify that the scope of work for the System/Sub-system is completed in accordance with the design, project specifications, project procedures, and any other contractual requirements.

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Tri-party Walk down, led by Contractor, shall involve Contractor's & Company's commissioning, Construction and QA/QC representatives. Sub-contractor's Commissioning and QC representatives shall join on case-to-case basis,

Punch list items shall be classified according to their degree of impact on safety, equipment integrity, and subsequent scheduled activities. The assignment of punch list classification (A, B or C) shall be in mutual agreement between CONTRACTOR, SUB-CONTRACTOR, and COMPANY representatives.

Refer to *Completions, Acceptance and Handover Procedure (PF0009-SLB-CO-PRO-0001)* for details of punch-list Management

11.3 COMMISSIONING HSE PRINCIPLES

Sub-Contractor commissioning team shall adhere to the Health Safety Environmental guidelines and work practices and to the HSE program applicable for the project.

11.3.1 Commissioning Safety

- Task risk assessments and job safety analysis will be carried out prior to the commencement of work.
- Toolbox talks will be carried out for all new work to ensure the team of involved personnel are well informed on roles, responsibilities, hazards and preventive measures prior to commencing the work.
- Approved working procedures, including Safety/Process Safety considerations and method statements will be followed.
- Assessments of hazardous substances prior to handling (With correct PPE usage and correct disposal of waste products)
- PTW Area Authorities are dedicated to commissioning team, in accordance with OPF PTW Guidelines, as applicable Commissioning (upon Energization) and Operations (upon Hydrocarbons In) take over the PTW Authority role over Construction in the whole Facility or in designated Areas.
- Periodic Safety audits to ensure compliance.
- "STOP " policy is made clear to all personnel for any doubtful actions.
- LOTO procedures shall be followed, using adequate material, in coordination with PTW / Isolation Permits, as applicable
- Warning signs and barriers erected to segregate systems / areas under testing
- RIR, STOP card or equivalent Risk Reporting to be encouraged and where necessary further investigation to be conducted to identify root cause. Safety Rewards schemes may be implemented to encourage such HSE awareness.
- Adequate Personal Protection Equipment (PPE) to be used on site
- Define clear roles and responsibilities for all involved personnel and ensure necessary competence.
- Field Change Notification and Site Technical Enquiries, Software Change Control and Control of Instrumentation Inhibits, Overrides and Forced Signals.
- Blinds and spades list, keys for panels, valve interlocking keys status.

11.3.2 PTW system

An effective safe system of work (e.g., Permit to Work/PTW) will be employed for the duration of the mechanical completion and commissioning works. The company's PTW specifications will be taken into consideration while developing project specific PTW Procedure applicable for construction and commissioning phase.

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Sub-contractor will engage trained and competent personnel to request and sign for receipt of permits to work under Sub-contractor Permit to Work System.

The PTW system will be progressively implemented on construction area basis followed by commissioning area basis. The designation of a physical area as construction, commissioning or operation safety Area is as below:

- Construction Safety Area: Project PTW Procedure will be managed by contractor
- Commissioning Safety Area: Project PTW Procedure will be managed by sub-contractor
- Operation's Safety Area: Company PTW system/procedure will be implemented and managed by Company after RFSU state.

Company's PTW system (Shall be implemented One week before RFSU). Operational Safety designated areas will have a clear demarcation with access control, signs and barriers.

Project PTW Procedure shall be developed by project HSE Manager and shall be reviewed by Construction Manager and commissioning Manager of Contractor and Sub-Contractor.

11.3.3 SIMOPS

Simultaneous Operations (SIMOPS) involving Construction, Commissioning and Operations should be identified during planning as well site execution phase of the project.

Simultaneous operations (SIMOPS) assessments studies will be carried out to see if risks may arise from interactions to occur between major hazardous concurrent activities within Commissioning, and/or between Construction and Commissioning.

The purpose of the SIMOPS assessments is:

- to identify the additional levels of risk introduced by simultaneous operations.
- to assess the acceptability of such additional risk
- to identify risk reduction methods, by analysing findings and recommendations from the SIMOPS assessments
- to develop the project specific SIMOPS procedure.

11.4 COMMISSIONING METHODOLOGY

Commissioning activities are undertaken, after system pre-commissioning completion, in order to dynamically verify functionality of equipment and to ensure that sub-systems, areas or facilities, forming part of a system, are in accordance with design specifications and hence the ability to bring that system into operation safely.

The Commissioning phase occurs progressively, at System or group of Subsystem level, and it is verified using relevant commissioning ITR-B and Operational Test Procedure (OTP). OTP provides a clear and detailed breakdown of the commissioning activities to safely commission the system at equipment and components level. Each OTP includes a detailed check list that shall be signed off and tracked through CCMS system to certify and follow-up all activities until the RFSU certificate is issued.

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Sub-Contractor, with necessary support by Contractor (as defined in PF0009-SLB-CO-PRO-0001 Completions, Acceptance and Handover Procedure), will be responsible to execute commissioning of all systems and to bring it to “Ready for Hydrocarbon-In” and “Ready for start-up” status. Commissioning activities shall broadly include,

11.4.1 Testing Of Process and E & I Systems:

- Cause & Effect Testing
- ICSS sequence checks
- ESD functional checks
- F&G system checks
- Execution of **Operational Test Procedures at system level**. (Refer Annexure-2)

11.4.2 Piping Integrity Testing and Inspection

Integrity testing and inspection of piping Systems prior to introduction of hydrocarbons should include:

- Preparation of high-pressure leak testing loop including flange management, as applicable
- Random Boroscopy of critical piping system as per vendor's guidelines. E.g., Compressor suction / compressor casing, before final re-instatement for leak testing and inerting.
- Leak testing of hydrocarbon systems
- System inerting as applicable to the process systems.

11.4.3 Leak testing of hydrocarbon systems

All the hydrocarbon systems piping flanged and threaded joints shall be leak tested using nitrogen as a pressure test medium. Third party contractors shall be engaged to perform the leak testing. The following activities shall be done as part of preparation for leak testing.

- The Leak Testing procedure shall be developed by SLB. This procedure will be a project specification for third party vendors to carry out the leak testing activities.
- Technical Requisition for leak testing shall be prepared by SLB commissioning team.
- Creating the Leak test pack number and Leak Test boundary mark-up shall be under the scope of third-party vendor and shall be reviewed and approved by Commissioning and company representatives.
- **All** leak test packs numbers shall be uploaded into CCMS and will be assigned to relevant systems and sub-systems
- Leak testing will be the last activity as part of the systems commissioning.

11.5 SYSTEM SPECIFIC COMMISSIONING ACTIVITIES

Apart from usual loop checks, functional checks and dynamic testing, the following system requires specific testing and commissioning activities which may also require vendor specialist support or third party support.

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11.5.1 ICSS SAT (Site Acceptance Test)

ICSS SATs shall provide further field checks and ensure that the integrated process, control and safety protective systems functions correctly before starting the process and utility systems commissioning activities followed by hydrocarbons are introduced into the facility. Packaged equipment's control systems (UCP) shall be tested onsite prior to commencing the commissioning activities.

Preparation for SAT is most critical and will be coordinated by ICSS Vendor specialists and SLB Commissioning team to ensure that when packages are connected to the controls and safeguarding systems at site, the communication protocols and connectivity are fully compatible and tested. Followed by completion of ICSS SAT, field loop checking, and functional testing can be started at sub-system level.

11.5.2 Commissioning of Refrigeration system

Refrigeration systems involve Drying & vacuum pulling for Refrigerant piping followed by charging of refrigerant prior to start-up. Adequate vacuuming is the key to smooth start-up of Refrigeration unit. Other commissioning activities such as functional testing and executing OTP will be carried out. Vendor supplied vacuum pump or external vacuum pump shall be used for vacuum pulling of Refrigeration system.

11.5.3 Commissioning of Amine Gas sweetening unit

Amine Gas sweetening systems involve the following activities and are key to the smooth start-up of Amine Gas sweetening unit.

- Degreasing of Amine system (associated equipment and piping in Amine circuit) using 5% MDEA and DM water solution using system's Amine pumps.
- Filling of Amine system with MDEA and circulation as per vendor's IOM and in presence of vendor representative.

11.5.4 Commissioning of Thermal Oxidizer

Thermal Oxidizer is installed with internal lining of Refractories. These refractories are dried at vendor shop facilities, but these refractories can be affected by the moisture ingress during shipping and construction phase of the project and therefore require a further dry out before start-up.

The site Refractory dry out activity requires the early introduction of fuel gas for Burner commissioning to complete dry out activities, thereby fuel gas system and associated piping and supporting package systems need to be commissioned early.

Refractory dry out will be carried out as per vendor's procedure and final Inspection with Company prior to RFSU acceptance.

All loop checks, functional checks and leak testing of fuel gas piping need to be completed prior to starting refractory dry out activities. Early introduction of fuel gas will require Mini PSUR.

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11.5.5 Commissioning of Hot oil system

Hot oil systems involve filling and circulation of heating medium fluid (Therminol 55 or Therminol 66). Heat medium will be filled through closed drain drum, starting from filling the hot oil expansion vessel followed by hot oil circulation piping loop.

All loop checks, functional checks and leak testing of fuel gas piping need to be completed prior to starting. Hot oil circulation pump will require hot alignment prior to final start-up of the hot oil circulation pumps.

11.5.6 Commissioning and flushing of lube oil systems

Following Rotating equipment packages of the AGIBA facility are comprising of the forced lubrication system. Lubrication system piping including, Site built piping and external lube oil coolers are subject to cleaning and flushing with system pump or external filtration skid to achieve cleanliness of the lube oil recommended by equipment vendor.

- Inlet gas compressor package – A & B
- Overhead gas compressor package – A & B
- Sales gas compressor package – A & B
- Refrigeration Compressor package
- Gas Turbine Generator packages – 5 nos.

The Lube oil system flushing procedure will be developed by SLB in-line with the lube oil system flushing requirement recommended in compressor package vendors IOMs. Flushing loops will be marked up into relevant vendor P & IDs. It is preferred that cleanliness of the lube oil is witnessed and approved by respective package vendor representatives.

11.6 COMMISSIONING SEQUENCE

Sub-contractors shall develop commissioning sequences and will be used to input the integrated project schedule to develop commissioning schedules which will be linked with Mechanical completion of the system and sub-system.

In addition to integrating the commissioning sequence into the project schedule, it will also help project planners to create MC skyline on commissioning priority basis. Such MC skyline will be issued by project planner on weekly or bi-weekly basis to provide contractors with the construction and mechanical completion priority.

The sequence of the Commissioning of AGIBA project facility shall be grouped and commissioned as below.

Group-1 E-house (Sub-station) /CCR ready for field commissioning activities

- Pre-commissioning and commissioning of temporary power for testing and energizing e-house and control systems
- Pre-commissioning and testing of Electrical System (E-House – MV/LV), including earthing and UPS using temporary power.

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- Commissioning of HVAC of e-house and control building (Combined building)
- Site acceptance test and commissioning of ICSS systems
- Commissioning of e-house/CCR Fire and Gas detection and protection systems
- E-house/CCR Telecom-CCTV
- E-house/CCR completion and ready for operation
- MV-E-house will be commissioned after commissioning LV E-House. However, energization with MV power will be done after commissioning and start-up of at least one GTG

Group-2 Major commissioning activities

- Plant earthing and lighting systems (as per associated sub-system MC and energization requirements)
- Energize field electrical and instrumentation circuits as MC of respective sub-systems
- Carryout systems loop checks based on commissioning sequence and MC status.
- Plant CCTV / PAGA
- Basic Utilities systems as below
 - ✓ Open drain and closed drain
 - ✓ Instrument Air and Nitrogen package
 - ✓ Emergency diesel generator and Diesel system (tank and distribution piping)
 - ✓ Service water followed by Fire water, potable water, sanitary water and sewage treatment.
 - ✓ DM water system
- Plant F & G systems commissioned on sequential basis.
- Flare system (Flare header till associated sub-system boundary isolation)
- Chemical Injection system
- Other utilities system as per project scope e.g., Fuel Gas Package and FG system (Including applicable tie-in piping)

Group-3 Process Systems and export piping systems.

Followed by completing commissioning of all utilities, the process systems commissioning can be sequenced based on start-up priority. This covers mainly commissioning systems without the introduction of hydrocarbon and using innocuous fluid. (e.g. Diesel, Water) Completion of system commissioning is also dependent on readiness of system for final leak testing and inerting.

For more details, Refer to *PF0009-SLB-CO-PLN-0002_Commissioning Sequence Plan and Logics*

11.7 SYSTEM COMMISSIONING COMPLETION

System commissioning completions will involve the following activities,

- Complete system wise commissioning checks and tests using B-ITRs and update CCMS
- Executing operational test procedures at system level update the CCMS
- Carry out leak testing of the hydrocarbon system (Utility systems leak testing will be done through services tests after charging the utilities with respective service fluid.)
- Compilation of System commissioning completion dossier (for each system)
- Update CCMS with system commissioning completion and raise RFSU certificate for each system.

Refer to [Completions, Acceptance and Handover Procedure \(PF0009-SLB-CO-PRO-0001\)](#) for details of System completion dossier.

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11.8 Start-up, and Stable Operation

11.8.1 Ready For Start-Up of Hydrocarbon Systems (RFSU)

RFSU is considered as the point when all activities necessary to support the introduction of hydrocarbons, including all utility and process utility, safeguarding and shutdown systems have been commissioned and integrity verified. The following actions will be taken prior to the declaration of RFSU:

- System(s) are fully commissioned, and system dossiers are approved.
- Close out of all (or part, if/as mutually agreed with client) category -B punch list items.
- System/s are ready for PSUR
- RFSU certificate has been signed off.

Followed by the issuance of the RFSU certificate, the system/s will be ready to conduct PSUR.

11.8.2 Systems RFSU Verification

Prior to approval of RFSU certificate for a system or group of systems by company, the System commissioning completion status will be reviewed and verified through following steps,

- a. Review and approval of RFSU Dossiers
- b. Review of Cause-and-effect sheets to ensure completion and sign-off
- c. Ensure all safeguarding instruments are tested
- d. Physical verification of system line-up for start-up and introduction of hydrocarbon
- e. Review of lock open and lock closed valve register
- f. Review of Mechanical and Electrical Isolation Register
- g. [Review of ICSS, ESD/F&G block, force & override Register](#)

[On satisfactory review and verification of system/s for RFSU, the company will approve the RFSU certificate](#)

11.8.3 Pre-Start-Up Review (PSUR)

The objective of the PSUR is to uniformly and consistently ensure that key process safety elements and deliverables have been completed and verified by competent people ensuring that hydrocarbons can be safely introduced and effectively managed with an acceptable level of risk (ALARP).

PSUR shall verify the closure of the following key elements:

- PSUR will be carried out as per project PSUR procedure and check lists there-in.
- All documents associated with start-up and operations are in place and approved. (e.g., Start-up and shutdown procedures, Plant operating procedure manuals, etc)
- The Safety Critical Equipment (SCEs) are tested and verified before Start-Up.
- All Process Safety Management design and engineering standards are met as per project specifications and CONTRACT.

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On Successful completion of PSUR and closing of all actions, Sub-Contractor shall issue the Ready for hydrocarbon introduction (RFHCIN) certificate to Company for their signoff and introduce Hydrocarbon into the Facility.

11.8.4 Commissioning and Start-Up with hydrocarbon (HC)

Sub-Contractor will ensure readiness for the commissioning and start-up of the hydrocarbon systems and carry out commissioning activities.

a. Pre-requisites to carry out commissioning and start-up of the systems require fuel gas or hydrocarbon

- Obtain RFSU certificate for the systems to be introduced with fuel gas or hydrocarbon feed for commissioning and start-up.
- Ensure timely Closeout of PSUR findings and actions to enable smooth and safe introduction of fuel gas or hydrocarbon feed for partial start-up. (Gas Compressors hydrocarbon commissioning.)
- Preparation for start-up (e.g., line walking, valve line-ups, emergency response exercise).
- Ensure sub-contractor's Startup support team is well prepared and ready to Support till end of the performance testing.
- Any logic overrides or forces, temporary Systems or Equipment because of Pre-Commissioning activities at the time of Commissioning shall be duly approved and recorded.
- Ensure readiness of the Facility to Introduce Hydrocarbon (Non-hydrocarbon systems are fully operational and Hydrocarbon systems are commissioned as applicable).
- Testing of plant ESD systems.

b. Systems require early introduction of fuel Gas

The following systems will be required to commission using fuel gas, which requires to obtain fuel gas from the company's export pipeline tie-in point upstream of the metering package. Gas from the export line will be routed to fuel gas package through metering skid.

- Refractory dry-out of thermal oxidizer refractory lining
- Hot oil system commissioning
- Commissioning and start-up of two GTGs for generating the power for commissioning of MV E-House followed by commissioning of MV motors of Refrigeration compressor, Overhead gas compressors, Amine circulation pump and MOP compressor.

c. Systems require to introduce feed gas

Inlet gas compressors and sales Gas compressor commissioning and tuning with the feed gas will be done soon after the introduction of feed gas. Commissioning of these systems will be done as per relevant systems OTP's.

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11.8.5 Full start-up and stable operations

After completion of the commissioning systems with hydrocarbon, (e.g. gas compression systems), other systems will be started-up in sequence as per *start-up and shutdown manual* ([Doc No. PF0009-SLB-CO-MAN-0001](#))

11.9 Performance Test Run

Upon achieving stable operation of the plant by COMPANY operations team, with support by Sub-Contractor's commissioning and operations team & required vendor specialists, Sub-Contractor will initiate and supervise the performance test run as per approved performance test run procedure. Ready for Performance testing certificate shall be issued by company on confirming stable operation of the facility.

The objective of the Performance Test Run is to prove that the plant is ready and capable of continuous operation. Should the feedstock not be available the performance Test will be conducted with the available feedstock subject to meeting minimum turndown condition corresponding to the feed stock. To successfully pass this test, the plant should operate without interruption within an agreed stability and variability/variation rates, for the duration (e.g., 72 hours), design feed rate and specification specified in the approved performance test procedure.

In case the Performance Tests, for the duration stipulated in the project specification cannot be performed for reasons not solely attributable to the Sub-Contractor including, but not limited to, non-availability of feedstock in sufficient quality and quantity etc., then the parties may mutually agree and attempt to carry out the Performance Tests later and/or possibly on unit-by-unit basis at a reduced throughput.

Project specific performance test procedure ([PF0009-SLB-CO-PRO-0015](#)) shall be developed by process commissioning lead in consultation with process engineering lead in-view of considering the process guarantees in line with the *Process Performance Guarantee* ([PF0009-SLB-PR-WRN-0001](#))

12.0 COMPLETIONS AND COMMISSIONING RESOURCES MANAGEMENT

Sub-contractor envisages and will manage various commissioning resources such as SLB commissioning team, Petrojet's support manpower, equipment vendor's field service specialists, third party contractors, temporary equipment for commissioning and commissioning special tools and consumables as described below,

12.1 SUB-CONTRACTOR'S COMMISSIONING ORGANISATION

Sub-contractor Commissioning Manager will develop organization chart suitable for commissioning, start-up, and performance testing of the AGIBA facility. Company O & M team will be integrated with sub-contractor's commissioning team for their training and familiarization with the facility commissioning, operations and maintenance.

Commissioning team will be mobilized gradually from planning and preparation phase through mechanical completion and commissioning and start-up phase.

Contractor's commissioning support manpower will be mobilized as necessary

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Refer to Annexure-3 for typical commissioning org. which will be separately agreed with contractor and company.

12.2 Vendors and Third-Party Sub-Contractor Mobilization Plan

Sub-Contractor envisages field technical support and supervision from equipment supplier's field specialists during installation phase through commissioning, start-up and performance testing of the facility.

Project specific vendor mobilization plan shall be developed by commissioning manager in coordination with commissioning leads, in-line with the project construction and commissioning schedule. This shall also include third party services contractors for leak testing etc.

12.3 Commissioning third party site services / certification services contractors

Following third party services contractors are envisaged for AGIBA project

- Chemical cleaning of compressor suction lines (as per equipment vendor's instructions)
- Lube oil system flushing (as per equipment vendor's instructions)
- Nitrogen leak testing
- Degreasing and flushing of Amine System with MDEA solvents.
- Certification of load testing of fixed lifting devices (EOT cranes, Monorails etc)
- ATEX certification by third party Inspector or in-house ATEX certified Inspector, if/as applicable to the project QC requirement.
- Any other third-party certification as per project specifications

12.4 Temporary Utilities and Equipment for Commissioning

Sub-Contractor, in line with the priorities of completing commissioning and based on actual availability of permanent utilities at site, will mobilize temporary utilities to complete the commissioning of systems as per project schedule.

Following temporary utilities and temporary equipment envisaged for AGIBA project,

Diesel Generator:

The capacity of the diesel generator will be estimated based on load list of the equipment and systems to be commissioned with temporary power for Pre-commissioning of e-house, CCR and LERs. Pre-commissioning of e-house/control system, field lighting, field loop checks, and LV motor testing are envisaged to be commissioned by temporary power.

Temporary equipment

The following temporary equipment are envisaged for AGIBA project

- Water collection tanks and hoses
- AC Load bank for EDG/GTG testing / DC Load bank for UPS testing
- De-watering pumps

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Special tools and test equipment

- Special Electrical test equipment for testing and commissioning electrical equipment in e-house
- Loop checking instruments.
- General Mechanical, Electrical and Instrumentation tool kits
- F&G, H2S detectors calibration, test equipment

Detailed list of special tools and test equipment will be prepared by the commissioning coordinator.

Consumables

- Water for dynamic commissioning of pumps
- Diesel for dynamic commissioning of diesel engine driven FW pumps, EDG, third party commissioning special equipment and any pumps require diesel for dynamic commissioning.
- Nitrogen for inerting and charging the seal system of the pumps.
- Test Gas for the testing of F&G detectors
- Demineralized water

13.0 COMMISSIONING INTERFACE MANAGEMENT

The Commissioning Manager will ensure that interface information associated with commissioning of the facility is available from concerned parties by providing input to **the project Interface register**. Following commissioning interfaces are envisaged for this project.

Internal Interfaces

- Interface with Engineering for MOC, Field engineering changes,
- Temporary overrides/Inhibits during commissioning,
- Red line mark-up (post MC) and green line mark-up (Post Commissioning)
- Interfaces with construction on temporary isolation carried out for performing commissioning activities or visa versa.

External Interfaces

- Interface with and/or between Contractor and sub-contractor(s) during execution of mechanical completions and commissioning.
- Interface with COMPANY for start-up gas and early fuel gas tie-ins.
- Interface with COMPANY for line-up fuel gas for early requirement of fuel gas and start-up pressurization gas
- Line-up of MOP flare gas with existing plant
- Waste disposal interfaces with Company (including permits)
- Interface with COMPANY for Local Authorities permits (Environmental, Firefighting, etc)

14.0 ANNEXURES

14.1 ANNEXURE-1 COMMISSIONING, START-UP DELIVERABLES

Sr.	DESCRIPTION	SLB DOCUMENT NUMBER
1	Commissioning execution plan	PF0009-SLB-CO-PLN-0001

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2	Completion, acceptance and handover procedure	PF0009-SLB-CO-PRO-0001
3	CCMS implementation procedure	PF0009-SLB-CO-PRO-0002
4	Commissioning sequence and logic	PF0009-SLB-CO-PLN-0002
5	Systems and sub-systems mark-up and system list	PF0009-SLB-CO-LSL-0001
6	A and B ITRs catalogues and ITR allocation matrix	PF0009-SLB-CO-LSL-0002
7	Electrical equipment energization procedure	PF0009-SLB-CO-PRO-0004
8	Livening-up (LUN) procedure	PF0009-SLB-CO-PRO-0005
9	Lock-out tag-out (LOTO) procedures including mechanical and electrical isolations/de isolation	PF0009-SLB-CO-PRO-0006
10	Leak testing and inerting procedure	PF0009-SLB-CO-PRO-0010
11	Lube oil system flushing procedure and mark-up drawings	PF0009-SLB-CO-PRO-0011
12	ICSS, ESD/F & G block, force & override procedure and register	PF0009-SLB-CO-PRO-0012
13	Loop checks and logic check procedure	PF0009-SLB-CO-PRO-0013
14	First fill lubricants, chemicals and test fluids/gases list	PF0009-SLB-CO-LSL-0003
15	Equipment and systems preservation plan, procedures and register	PF0009-SLB-CO-PRO-0014
16	Performance test procedure	PF0009-SLB-CO-PRO-0015
17	Start-up and shutdown manual	PF0009-SLB-CO-MAN-0001
18	Plant operating procedure manual	PF0009-SLB-CO-MAN-0002

14.2 ANNEXURE-2 LIST OF OPERATIONAL TEST PROCEDURES (OTP)

Sr.	DESCRIPTION OF OPERATIONAL TEST PROCEDURE
1	Operational test procedure for inlet manifold and separator
2	Operational test procedure for inlet gas compression train 1
3	Operational test procedure for inlet gas compression train 2
4	Operational test procedure for mercury removal unit (MRU)
5	Operational test procedure for gas sweetening unit (GSU)
6	Operational test procedure for amine drains drum & pumps (GSU)
7	Operational test procedure for gas dehydration unit (GDU)
8	Operational test procedure for glycol drains drum & pumps
9	Operational test procedure for dew point control system
10	Operational test procedure for refrigerant compressor-A
11	Operational test procedure for refrigerant compressor-B
12	Operational test procedure for refrigerant compressor-C
13	Operational test procedure for overhead gas compression system (train 1)
14	Operational test procedure for overhead gas compression system (train 2)

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15	Operational test procedure for sales gas compression system (train 1)
16	Operational test procedure for sales gas compression system (train 2)
17	Operational test procedure for mop flare gas compression system
18	Operational test procedure for sales gas metering system
19	Operational test procedure for condensate stabilization system
20	Operational test procedure for corrosion inhibitor system
21	Operational test procedure for emulsifier injection system
22	Operational test procedure for antifoam injection system
23	Operational test procedure for scale inhibitor system
24	Operational test procedure for meg injection system
25	Operational test procedure for sodium hypochlorite system
26	Operational test procedure for hot oil system
27	Operational test procedure for hp flare system
28	Operational test procedure for lp flare system
29	Operational test procedure for thermal oxidizer system
30	Operational test procedure for fuel gas system
31	Operational test procedure for raw water system
32	Operational test procedure for service water system
33	Operational test procedure for demineralized water system
34	Operational test procedure for potable water system
35	Operational test procedure for closed drain system
36	Operational test procedure for open drain system
37	Operational test procedure for instrument air compressor 1
38	Operational test procedure for instrument air compressor 2
39	Operational test procedure for instrument air compressor 3
40	Operational test procedure for instrument / utility air system
41	Operational test procedure for nitrogen system
42	Operational test procedure for fire water systems
43	Operational test procedure for LV systems - LV e-house
44	Operational test procedure for LV systems - MV-e-house
45	Operational test procedure for LV systems - CCR

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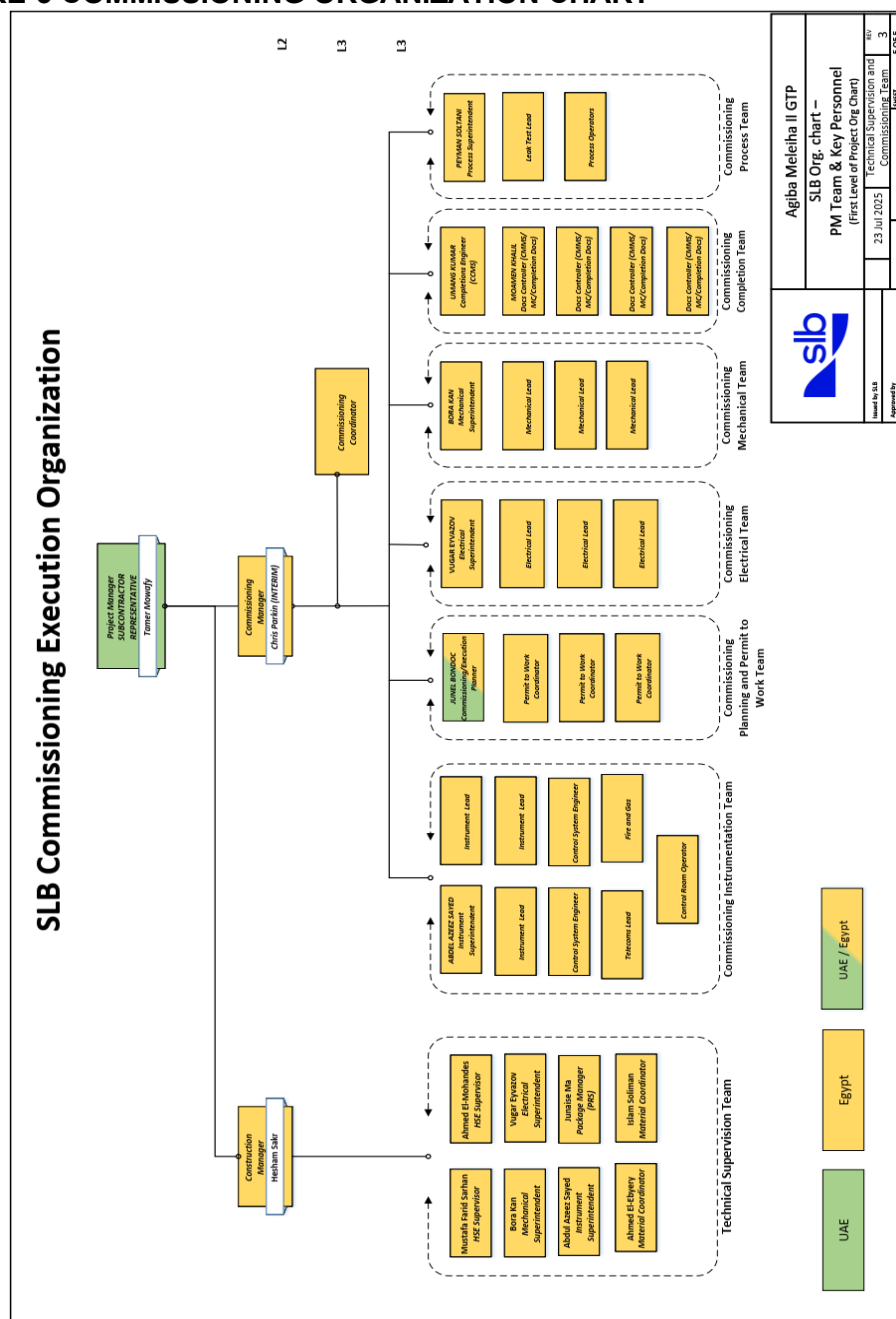
46	Operational test procedure for LV systems - plant area
47	Operational test procedure for heat tracing
48	Operational test procedure for cathodic protection
49	Operational test procedure for AC UPS
50	Operational test procedure for MV-e-house UPS DB (building)
51	Operational test procedure for CCR building – UPS DB
52	Operational test procedure for emergency power EDG-1
53	Operational test procedure for emergency power EDG-2
54	Operational test procedure for diesel tank and distribution
55	Operational test procedure for LV e-house and emergency LV switchboard
56	Operational test procedure for LV e-house & main emergency lighting switchboard
57	Operational test procedure for LV e-house- GTG MCC
58	Operational test procedure for LV system
59	Operational test procedure for MV system
60	Operational test procedure for earthing systems
61	Operational test procedure for GTG-A
62	Operational test procedure for GTG -B
63	Operational test procedure for GTG -C
64	Operational test procedure for GTG -D
65	Operational test procedure for GTG -E
66	Operational test procedure for PCS control system
67	Operational test procedure for ESD control system
68	Operational test procedure for ESD / C & E effect - process and utility area
69	Operational test procedure for ESD / C & E effect - third party packages
70	Operational test procedure for FGS control system
71	Operational test procedure for LV e-house F & G system
72	Operational test procedure for MV e-house F & G system
73	Operational test procedure for CCR F & G system
74	Operational test procedure for fire & gas system fire zone-00
75	Operational test procedure for fire & gas system fire zone-03
76	Operational test procedure for fire & gas system fire zone-04

		N/A	Validity Status	Revision Number
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77	Operational test procedure for fire & gas system fire zone-05
78	Operational test procedure for fire & gas system fire zone-06
79	Operational test procedure for fire & gas system fire zone-08
80	Operational test procedure for fire & gas system fire zone-09
81	Operational test procedure for fire & gas system fire zone-24
82	Operational test procedure for fire & gas system fire zone-01
83	Operational test procedure for fire & gas system fire zone-02
84	Operational test procedure for fire & gas system fire zone-07
85	Operational test procedure for fire & gas system fire zone-10
86	Operational test procedure for fire & gas system fire zone-11
87	Operational test procedure for fire & gas system fire zone-12
88	Operational test procedure for fire & gas system for building areas
89	Operational test procedure for LV e-house telecom system
90	Operational test procedure for MV e-house telecom system
91	Operational test procedure for CCR telecom system
92	Operational test procedure for process area telecom system
93	Operational test procedure for utility area telecom system
94	Operational test procedure for LV e-house fire fighting
95	Operational test procedure for MV e-house fire fighting
96	Operational test procedure for CCR building fire fighting
97	Operational test procedure for process & utility area fire fighting
98	Operational test procedure for LV e-house HVAC
99	Operational test procedure for MV e-house HVAC
100	Operational test procedure for CCR HVAC

 		N/A	Validity Status	Revision Number
			EX-DE	04
Company Document ID	Contractor Document ID	Vendor Document ID	Sheet of Sheets	
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14.3 ANNEXURE-3 COMMISSIONING ORGANIZATION CHART



Note: Above Commissioning Organization is extracted from the project overall Organization chart and represented here for reference. Names and mobilization date to be referred from main project Organization chart.



Project Name

Facility/Subfacility name

Facility/Subfacility code

Country

MELEIHA PHASE II – NEW GTP (EPF)

Melehia Gas Plant

3134

EGYPT

Transmittal out ID

OUT-3134-A9U-PTJ-00553

Date21-10-2025

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ID	COMPANY ID	CONTRACTOR ID	Title	Revision Number	Validity status	ID	Date	Send Reason	Send Reason Out		
1	3134-00-D-W-PB-13235	PF0009-SLB-CO-PLN-0001	COMMISSIONING EXECUTION PLAN	03	EX-DE	IN-3134-PTJ-A9U-06188	14-10-2025	A	Approved		

Related documents

Document	Title
DOC_COMMENT_3134-00-D-W-PB-13235_EDE03_PTJ	Comment document belonging to document 3134-00-D-W-PB-13235_EDE03

Raised by: Ahmed Abdelsalam

Project document control